Significance Statement. Most protein families have fewer than 100 known members, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled exactly via Langevin dynamics, with no training, no GPU, and no external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and fold into the correct three-dimensional structure across seven diverse Pfam families. A simple linear relationship between PCA dimensionality and the critical sampling temperature enables fully automatic operation from a seed alignment, opening training-free sequence generation to the vast majority of protein families that are too small for conventional deep learning.

Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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Abstract

Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models on thousands to millions of examples. Yet most protein families are small: the median Pfam family has fewer than 100 members, a regime where learned models overfit or collapse. We propose *stochastic attention*, a training-free sampler that treats the modern Hopfield energy over a set of stored protein sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in closed form by the residual of a single softmax attention operation, eliminating the need for a trained score network, external pretraining data, or GPU resources. Each iteration costs $\mathcal{O}(dK)$ — two matrix-vector products. Across seven Pfam families (K=37–420 sequences, L=23–262 residues), stochastic attention generates sequences with low amino acid composition divergence (KL < 0.06), substantial novelty (0.42–0.65), and structural plausibility confirmed by ESMFold and AlphaFold2 (mean pLDDT up to 95). Strikingly, generated sequences achieve significantly *higher* TM-scores to canonical family folds than natural family members in five of seven families ($p < 0.001$). Head-to-head comparison with profile HMMs, EvoDiff, and the MSA Transformer reveals a critical distinction: learned baselines produce sequences that drift far from the family (as low as 11% sequence identity), while stochastic attention maintains 51–61% identity — staying within the functional envelope while remaining genuinely novel. Stochastic attention achieves this in seconds on a laptop, compared to hours of GPU or CPU time for pretrained alternatives, and scales stably down to K=20 sequences. The critical temperature governing the generation–retrieval transition is predicted from PCA dimensionality alone (R^2=0.97), enabling fully automatic operation from a seed alignment.

1 Introduction

The protein folding problem, predicting three-dimensional structure from amino acid sequence, stood as one of the grand challenges in molecular biology for half a century. AlphaFold2 [1] and

RoseTTAFold [2] effectively solved it, achieving experimental-level accuracy across the proteome and earning the 2024 Nobel Prize in Chemistry. With structure prediction largely in hand, the frontier has shifted decisively toward the inverse problem: *designing* novel protein sequences with desired properties [3, 4]. This shift has catalyzed a wave of generative models for protein design, spanning structure-conditioned methods, sequence-level language models, and diffusion-based approaches.

On the sequence side, profile hidden Markov models (pHMMs) [16, 17] have long captured positional emission and transition probabilities from multiple sequence alignments (MSAs), but they model only single-site statistics and miss the pairwise and higher-order couplings critical for maintaining tertiary structure. Direct coupling analysis (DCA) and Potts models [18, 19] capture pairwise epistasis, and Trinquier et al. [20] showed that autoregressive models over DCA-derived distributions can generate realistic sequences efficiently. Deep generative models have pushed further: DeepSequence [21] learns latent representations via a VAE; ProtGPT2 [8] and ProGen [7] generate sequences autoregressively from large language models pretrained on UniRef50; the MSA Transformer [9] processes entire alignments via tied row attention; EvoDiff [10] applies discrete diffusion to MSA-conditioned generation; and Tranception [22] combines autoregressive transformers with inference-time retrieval. On the structure side, ProteinMPNN [5] designs sequences conditioned on backbone coordinates via message-passing, RFdiffusion [6] generates novel backbones through denoising diffusion, and inverse folding models [26, 24] learn sequence–structure mappings from millions of examples. These structure-based methods are powerful when a target backbone is available but do not address the problem we consider: generating novel sequences from a family alignment alone.

All of these approaches share a common requirement: large training corpora. Structure-conditioned models need backbone templates; sequence-level models are typically pretrained on millions of sequences and then fine-tuned or conditioned on family-specific alignments. This creates a blind spot. The median Pfam family contains fewer than 100 full-length members [11], and the long tail of protein space (orphan families, newly discovered clades, engineered libraries) has even less data. In this scarce-data regime, learned generative models overfit, collapse in diversity, or fail to train altogether. Yet these small families are precisely where generative tools could have the greatest impact, by expanding sparse natural repertoires for functional characterization or directed evolution [12].

A separate line of work connects attention mechanisms to energy-based models. Ramsauer et al. [13] showed that transformer-style attention is equivalent to one step of the retrieval dynamics on the modern Hopfield energy with a log-sum-exp interaction function, building on the dense associative memory framework of Krotov and Hopfield [27]. Concurrently, score-based generative models [30, 31] and denoising diffusion models [32] have demonstrated that samples can be drawn from complex distributions by learning a neural score function $\nabla \log p(\mathbf{x})$ and running Langevin dynamics. These methods achieve state-of-the-art sample quality but require a trained score network. The Energy Transformer [28] exploits the Hopfield energy for discriminative tasks but does not sample from the associated Boltzmann distribution.

We propose a fundamentally different approach that requires no training at all. We observe that the modern Hopfield energy over a set of stored protein sequences defines a Boltzmann distribution whose score function is given in closed form by the residual of a single attention operation: $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}$ is the softmax attention map. This eliminates score-estimation error and permits convergence guarantees inherited from the ULA literature [33, 34]. Langevin dynamics on this energy yields a training-free sampler that we call *stochastic attention*: each iteration performs a softmax-weighted pull toward stored family members, a contraction, and an isotropic Gaussian perturbation, requiring only two matrix-vector products per step ($\mathcal{O}(dK)$ cost, no GPU, no backpropagation). Our prior work [29] introduced stochastic attention and validated it on synthetic data, MNIST digits, financial time series, and a single protein family (RRM). The present work extends this framework to the protein domain with multiple families, structure-level validation, scaling analysis, and protein-specific baselines.

The key question this paper addresses is: *what can you generate from a small protein family alignment with no training?* Across seven Pfam families spanning a ten-fold range of family sizes (K=37–420) and a seven-fold range of sequence lengths (L=23–262), we show that stochastic attention produces novel sequences that preserve family-level amino acid composition (KL < 0.06) and achieve substantial novelty (0.42–0.65 in PCA space), while folding into the correct three-dimensional architecture as confirmed by ESMFold [14] structure prediction and TM-align [15] structural comparison. We characterize the critical inverse temperature β^* at which the sampler transitions from diffuse exploration to pattern retrieval, and show that β^* can be predicted from PCA dimensionality alone (R^2=0.97), enabling fully automatic operation from a seed alignment. A systematic scaling study reveals stable generation quality down to K=20 sequences. Head-to-head comparison with profile HMMs [16], EvoDiff [10], and the MSA Transformer [9] reveals that stochastic attention uniquely balances compositional fidelity, sequence novelty, and structural viability without external training data.

2 Theory

2.1 From Hopfield Energy to Score Function

Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions (after encoding and, optionally, PCA projection). The modern Hopfield energy [13] is

$$E(\boldsymbol{\xi}) = \tfrac{1}{2}\|\boldsymbol{\xi}\|^2 - \tfrac{1}{\beta} \log \sum_{k=1}^{K} \exp\bigl(\beta\, \mathbf{m}_k^\top \boldsymbol{\xi}\bigr), \tag{1}$$

where $\beta > 0$ is an inverse temperature. The gradient is $\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \mathbf{T}(\boldsymbol{\xi})$, where

$$\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X}\ \mathrm{softmax}\bigl(\beta\, \mathbf{X}^\top \boldsymbol{\xi}\bigr) \tag{2}$$

is the softmax attention retrieval map [13]. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta\, E(\boldsymbol{\xi}))$ has score function

$$\nabla \log p_\beta(\boldsymbol{\xi}) = -\beta\, \nabla E(\boldsymbol{\xi}) = \beta\bigl(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi}\bigr). \tag{3}$$

This score is *exact* and computed by a single attention operation; no score network or training is required.

Applying the unadjusted Langevin algorithm (ULA) [35] to the potential $U = \beta E$ with step size $\tilde{\alpha} = \alpha/\beta$ yields the *stochastic attention* update:

$$\boxed{\boldsymbol{\xi}_{t+1} = (1-\alpha)\, \boldsymbol{\xi}_t + \alpha\, \mathbf{X}\ \mathrm{softmax}\bigl(\beta\, \mathbf{X}^\top \boldsymbol{\xi}_t\bigr) + \sqrt{\tfrac{2\alpha}{\beta}}\, \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \tag{4}$$

Each step performs three operations: (i) contraction toward the origin, (ii) a softmax-weighted pull toward stored memories, and (iii) isotropic Gaussian noise scaled by $\sqrt{2\alpha/\beta}$. The per-step cost is $\mathcal{O}(dK)$: two matrix-vector products and one softmax. As $\beta \to \infty$ the noise vanishes and the update reduces to deterministic retrieval; as $\beta \to 0$ sampling becomes noise-dominated.

2.2 Critical Temperature from PCA Dimensionality

The attention entropy $H(\beta) = -\sum_k p_k \log p_k$, where $p_k = \mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi})_k$, quantifies selectivity: $H = \log K$ when attention is uniform (generation) and $H \to 0$ when attention concentrates on a single memory (retrieval). The entropy derivative satisfies [29]

$$\frac{dH}{d\beta} = -\beta\, \mathrm{Var}_{\mathbf{p}}(e), \tag{5}$$

where $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ are the query-memory similarities. The maximum rate of entropy loss occurs at the inflection point β^* satisfying $d^2H/d\beta^2 = 0$, which marks the boundary between the generation and retrieval regimes. The form of β^* is constrained by the geometry of the stored patterns. Because the memory columns lie on the unit sphere $\mathbb{S}^{d-1}$, concentration of measure dictates that the similarity variance is $\mathrm{Var}(e_k) = 1/d$ exactly for any query drawn uniformly from $\mathbb{S}^{d-1}$ (Appendix S1.2.1). The characteristic similarity scale is therefore $\sigma = 1/\sqrt{d}$, and since β enters the softmax as βe_k, the inflection must occur when $\beta\sigma = \mathcal{O}(1)$, giving $\beta^* \sim \sqrt{d}$. A separate calculation shows that the softmax entropy over K i.i.d. Gaussian scores undergoes its inflection at a dimensionless temperature $\tau^* \approx 1.6$ that is nearly independent of K for $K \geq 30$ (Appendix S1.2.1), providing a universal baseline for the transition.

Empirically, we find a simple linear relationship across seven Pfam families and WW-domain subsamples spanning $K = 20$–420 and $d = 18$–186 (Section 5):

$$\beta^* \approx 1.57 + 0.28\sqrt{d}, \tag{6}$$

which captures 96% of the variance ($R^2 = 0.962$). The intercept $1.57 \approx \tau^*$ recovers the universal softmax inflection baseline, confirming that the same dimensionless transition governs both idealized Gaussian scores and real protein similarities. The coefficient 0.28 of $\sqrt{d} = 1/\sigma$ is substantially smaller than the pure Gaussian prediction $\tau^*/\sigma \approx 1.6\sqrt{d}$; this reduction arises because β^* is computed at stored patterns $\mathbf{m}_k$, where the self-similarity $\mathbf{m}_k^\top \mathbf{m}_k = 1$ anchors one score at the maximum of $\mathbb{S}^{d-1}$ and makes concentration easier than for a random query (Appendix S1.2.2). From a practical standpoint, Eq. (6) enables fully automatic operation: given a seed alignment, d is determined by PCA at 95% variance, and β^* follows without a temperature sweep.

2.3 Regularity and Convergence

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\max}^2/2$, where $\sigma_{\max} = \|\mathbf{X}\|_{\text{op}}$, and satisfies the dissipativity condition $\langle \nabla E(\boldsymbol{\xi}), \boldsymbol{\xi} \rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$ [29]. When $\beta\sigma_{\max}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [34, 33].

In the protein setting, operating temperatures of interest satisfy $\beta\sigma_{\max}^2 \gg 2$, placing the sampler in a non-convex regime where the energy landscape has multiple metastable basins concentrated near the stored patterns. Following standard practice for multimodal Langevin sampling [36], we adopt a multi-chain protocol: N_c independent chains are initialized near distinct stored patterns, exploiting fast intra-basin mixing while achieving inter-basin coverage through diverse initialization. The Metropolis-adjusted variant (MALA) provides an acceptance-rate diagnostic: acceptance rates near 1.0 confirm that ULA discretization bias is negligible for the chosen step size α.

3 Experiments

We applied stochastic attention to protein sequence generation across seven Pfam families and conducted a systematic scaling study to characterize performance as a function of family size. All experiments used the unadjusted Langevin algorithm (ULA) with step size $\alpha = 0.01$. For each family and condition, we ran 30 independent chains of $T = 5{,}000$ iterations, initialized near randomly chosen stored patterns with Gaussian perturbation ($\sigma_{\text{init}} = 0.01$). After discarding 2,000 burn-in iterations and thinning every 100 steps, we retained 5 evenly spaced samples per chain for a total of $S = 150$ generated sequences per condition. Metropolis-adjusted Langevin algorithm (MALA) chains were run in parallel with identical parameters to provide acceptance-rate diagnostics. Throughout, we used PCA dimensionality reduction retaining 95% of the variance in the one-hot encoded alignment, followed by unit-norm projection to form the memory matrix $\hat{\mathbf{X}}$.

The inverse temperature was set automatically for each family via the entropy inflection criterion (Section 2.2). We identified β^* from the attention entropy curve and operated in two regimes: a *generation* regime at $\beta_{\text{gen}} \approx 2\beta^*$, where the sampler explores broadly while respecting family statistics, and a *retrieval* regime at $\beta_{\text{ret}} \approx 20\beta^*$, where samples remain close to stored patterns. Three simple baselines were evaluated under identical conditions: bootstrap replay (uniform resampling of stored patterns), Gaussian perturbation (stored pattern plus isotropic noise matched to the SA noise scale), and random convex combination (Dirichlet-weighted average of all stored patterns).

We evaluated generated sequences using five metrics. *Novelty* measured departure from stored patterns as $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space. *Diversity* was the mean pairwise cosine distance among generated samples. *Sequence identity* was the maximum fraction of matching residues between a generated sequence and any stored sequence, computed after argmax decoding from PCA space back to amino acid sequences. *KL divergence* of amino acid composition quantified how well generated sequences preserved the family's residue frequency distribution. *Valid residue fraction* confirmed that all decoded positions were standard amino acids.

Protein families and scaling study. We selected seven families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, K=68, L=71), SH3 (PF00018, K=55,

L=48), WW (PF00397, K=420, L=31), Kunitz (PF00014, K=99, L=53), zinc finger C2H2 (PF00096, K=151, L=23), PDZ (PF00595, K=44, L=83), and protein kinase (PF00069, K=37, L=262). Seed alignments were downloaded from InterPro [11] and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. The resulting PCA dimensions ranged from d=34 (Pkinase) to d=186 (WW). To characterize how generation quality degrades with decreasing family size, we also conducted a scaling study in which we fixed the WW domain (the largest family, K=420) and subsampled to $K \in \{20, 50, 100, 200, 400\}$. At each value of K we drew five independent random subsets, rebuilt the memory matrix from scratch (re-computing PCA and β^*), ran stochastic attention and all baselines, and recorded metrics; this design isolates the effect of family size from family identity, and the five repeats provide error estimates.

Learned baselines. To contextualize SA against methods that require pretraining on large external corpora, we compared three established generative approaches for protein sequences. (i) *Profile HMM emit*: we built a profile HMM from each family's seed alignment using HMMER3 [16] (`hmmbuild -amino`) and emitted S=150 sequences with `hmmemit -N 150 -seed 42`. Emitted sequences were filtered to standard amino acids and truncated or padded to the alignment length L. (ii) *EvoDiff*: we used the MSA-conditioned order-agnostic discrete diffusion model (MSA_OA_DM_RANDSUB) [10], which was pretrained on UniRef50 MSAs. Generation used the default denoising schedule with MaxHamming subsampling of the conditioning MSA (depth ≤ 64). (iii) *MSA Transformer*: we used the 100M-parameter MSA Transformer [9] (esm_msa1b_t12_100M_UR50S) for generation via iterative Gibbs-like masked language model sampling: a fully masked query row was appended to the family MSA (depth ≤ 128), and 50 rounds of masking 15% of positions and resampling from the softmax were performed. For each baseline, the same 150-sequence budget and the same evaluation metrics (KL, novelty, sequence identity) were applied. EvoDiff was run on CPU; for the Pkinase family (L=262), a single sequence required $\sim$7 minutes, yielding an estimated $\sim$18 hours for 150 sequences (compared to seconds for SA), so this family was excluded from the EvoDiff comparison.

Structure validation. To assess whether generated sequences adopt structures consistent with their target family, we predicted the three-dimensional structure of 50 sequences per method per family using the ESMFold API [14]. For each predicted structure, we extracted the mean predicted local distance difference test (pLDDT) score from the B-factor column as a measure of folding confidence (pLDDT > 70 indicates a confident prediction). We then compared each predicted structure to a representative experimentally determined structure for the family using TM-align [15] (TM-score > 0.5 indicates the same fold). Reference structures were obtained from the RCSB PDB: 1FXL (RRM), 1SHG (SH3), 1PIN (WW), 1BPI (Kunitz), 1ZAA (zf-C2H2), 1BE9 (PDZ), and 1ATP (Pkinase). As a positive control, we also predicted structures for 50 stored (natural) sequences from each family, providing an upper bound on expected pLDDT and TM-score. As an independent cross-validation, we repeated the structure prediction for all seven families using AlphaFold2 [1] via ColabFold [37], predicting 50 SA-generated and 50 stored sequences per family with the `alphafold2_ptm` model (single model, 3 recycles, no relaxation). For each family, we selected the SA-generated structure with the highest mean pLDDT and superimposed it on the experimental reference using PyMOL to produce the structural gallery in Fig. 6.

Critical temperature prediction. We pooled n=32 data points (the 7 Pfam families and 25 WW scaling replicates, 5 family sizes $\times$ 5 repeats) and fit ordinary least-squares regression models predicting β^* from MSA-derived features. We considered four candidate features: $\sqrt{d}$ (PCA dimensionality at 95% retained variance), mean per-column Shannon entropy $\bar{H}_{\mathrm{col}}$, $\log K_{\mathrm{eff}}$ (effective number of sequences at 80% identity), and spectral concentration $\lambda_1/\mathrm{tr}(\mathbf{C})$ (leading eigenvalue of the sequence covariance, normalized by trace). We evaluated nested models by adding features in order of marginal R^2 improvement. Uncertainty on regression coefficients was quantified via nonparametric bootstrap: we resampled the 32 observations with replacement 10,000 times, refitting the model on each resample, and reported the standard deviation of the bootstrap distribution as the coefficient SE. Model comparison used root-mean-square error (RMSE) and R^2, with the naive random-pattern prediction $\beta^* = \sqrt{d}$ as a reference baseline.

4 Results

We ran stochastic attention on all seven Pfam families and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Table 2, Fig. 1).

We first examined whether the sampler converged to the correct Boltzmann distribution by comparing ULA and MALA outputs. The MALA acceptance rates exceeded 99.6% in every family tested (range: 99.6%–99.9%), confirming that the ULA discretization bias was negligible at α=0.01. The two samplers produced near-identical metrics: for example, in the RRM family, SA retrieval achieved novelty 0.243 and KL divergence 0.107, while MALA yielded 0.249 and 0.112, respectively. This agreement held across all seven families, which justified the use of the computationally cheaper ULA in subsequent experiments.

We next assessed amino acid composition fidelity across families. In the generation regime ($\beta \approx 2\beta^*$), stochastic attention achieved KL divergences below 0.06 in every family, with several families below 0.02: WW achieved KL=0.008, Kunitz 0.013, and zinc finger 0.013. These values were substantially lower than the bootstrap baseline (KL range: 0.007–0.133), despite the fact that bootstrap samples are exact copies of stored patterns and thus trivially preserve composition at the individual-sequence level. The improvement arose because SA generation samples from the Boltzmann distribution over the entire family, producing a more representative ensemble than uniform resampling of a finite set. The convex combination baseline performed poorly, with KL divergences ranging from 0.06 (Pkinase) to 2.35 (zinc finger), which indicated that averaging across all stored patterns destroyed the peaked, position-specific residue distributions characteristic of conserved protein families (SI Appendix, Fig. S1).

We then examined whether the generated sequences were genuinely novel or merely replaying stored patterns. In the generation regime, novelty ranged from 0.42 (PDZ) to 0.65 (WW), indicating substantial departure from any single stored memory (Table 2). By contrast, bootstrap samples had zero novelty by construction, and Gaussian perturbation produced negligible novelty (< 0.02 in all families), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern. The retrieval regime produced intermediate novelty (0.17–0.35), consistent with samples that explore locally around stored patterns without fully escaping their basins. Nearest sequence identity in amino acid space confirmed this picture: generation-regime sequences had 52–61% identity to the nearest stored sequence, comparable to the typical within-family identity, while retrieval-regime sequences remained at 53–85% identity depending on the family's intrinsic diversity.

We also characterized how the critical temperature varied across families. The empirical β^* ranged from 3.2 (PDZ, Pkinase) to 5.4 (WW), consistently lower than the $\sqrt{d}$ prediction for random patterns (5.8–13.6). The ratio $\beta^*/\sqrt{d}$ was approximately 0.4–0.6 across all families, which indicated that the structured similarity spectrum of real protein sequences, arising from conserved positions, correlated substitutions, and phylogenetic signal, shifts the retrieval-to-generation transition to lower temperatures. Families with higher mean column entropy ($\bar{H}_{\text{col}}$) tended to have lower $\beta^*/\sqrt{d}$ ratios, consistent with the intuition that more diverse families require less thermal energy to escape stored patterns. The effective number of sequences K_{eff}, computed by reweighting at 80% identity, equaled K in all but the WW family (K_{eff}=419 vs. K=420), indicating minimal redundancy in the Pfam seed alignments.

We next turned to the scaling study, in which we subsampled the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. 2). We observed that stochastic attention in the generation regime maintained low KL divergence across the entire range: KL=0.019 $\pm$ 0.008 at K=20, improving monotonically to KL=0.010 $\pm$ 0.002 at K=400. This graceful degradation contrasted sharply with the convex combination baseline, whose KL divergence exceeded 0.8 at every family size and worsened with increasing K (0.81 at K=20 to 1.50 at K=400), confirming that averaging over more patterns compounds the smoothing artifact. Bootstrap and Gaussian perturbation maintained moderate KL values (0.02–0.03) but produced no novelty at any K, making them ineffective as generative methods.

The scaling study also revealed how PCA dimensionality and the critical temperature co-varied with family size. At K=20 the PCA reduction retained only d=17–18 components, producing a compact latent space in which β^*=2.7; at K=400 the richer alignment supported d=182–183 components and β^*=5.5. Despite this six-fold increase in effective dimensionality, SA generation quality remained stable, which suggested that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 at K=20 to 0.74 at K=400), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape.

We finally asked whether the critical temperature could be predicted from alignment statistics alone, without running an explicit temperature sweep (Fig. 3). We pooled 32 data points (the 7 Pfam families and 25 WW scaling replicates) and regressed β^* against MSA-derived features. We found that a simple two-parameter model, $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (intercept 1.52 ± 0.08, slope 0.282 ± 0.007; $\pm$ denotes bootstrap SE, 10,000 resamples), explained 97% of the variance (R^2=0.969, RMSE=0.16), with d the PCA dimension at 95% retained variance. Adding further predictors (mean column entropy $\bar{H}_{\mathrm{col}}$, effective number of sequences K_{eff}, or spectral concentration $\lambda_1/\mathrm{tr}(\mathbf{C})$) produced negligible improvement ($\Delta R^2 < 0.001$), which indicated that PCA dimensionality captured most of the relevant variation. The fitted slope of 0.28 was roughly half the $\sqrt{d}$ coefficient predicted for random patterns, consistent with the empirical ratio $\beta^*/\sqrt{d} \approx 0.4$–$0.6$ observed across families. Importantly, the naive random-pattern prediction $\beta^* = \sqrt{d}$ yielded RMSE=5.3, a 33× larger error, confirming that the structured similarity spectrum of real protein families shifts the phase transition to substantially lower temperatures. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a temperature sweep.

We compared stochastic attention against three established learned baselines: profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs MLM sampling) (Fig. 4). We used Wilcoxon rank-sum tests on per-chain estimates (30 chains × 5 samples) with Cohen's d effect sizes. EvoDiff achieved the lowest KL divergence in most families (0.003–0.011), consistent with its pretraining on millions of UniRef50 MSAs, and profile HMM emit also produced low KL values (0.006–0.030) by directly capturing position-specific residue frequencies. Stochastic attention generation was competitive with both (0.008–0.060) despite using only the family's seed alignment, typically 37–420 sequences, with no external training data.

However, the three metrics together revealed a trade-off that no single learned baseline resolved. The critical metric was nearest sequence identity, where all three baselines differed significantly from SA generation ($p < 0.001$ in all families). Profile HMM emit produced the highest novelty (0.56–0.71) but the lowest sequence identity to stored patterns (11–42%; SA gen vs. HMM emit: $d = 3.6$–24.6 across families), indicating sequences so divergent from the family that they likely fall outside its functional envelope. The MSA Transformer showed a similar pattern: high novelty (0.37–0.67) but low sequence identity (9–48%; $d = 2.1$–26.8). EvoDiff balanced KL and novelty more effectively but still produced significantly lower sequence identity than SA generation in every family tested ($p < 0.001$, $d = 1.6$–5.5), and its compute cost scaled prohibitively with sequence length: generating 150 sequences for the Pkinase family (L=262) required an estimated 18 hours on CPU, compared to seconds for SA. Stochastic attention generation uniquely occupied the regime of low KL divergence, substantial novelty (0.42–0.65), and moderate sequence identity (51–61%), generating sequences that departed meaningfully from stored patterns while remaining within the family's characteristic sequence space. Critically, SA achieved this balance without pretraining, backpropagation, or GPU resources, using only the information present in the seed alignment itself.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 5), comparing SA generation to stored (natural) sequences via Wilcoxon rank-sum tests with Cohen's d effect sizes. SA-generated sequences achieved mean pLDDT scores statistically indistinguishable from stored natural sequences in five of seven families (Wilcoxon $p > 0.05$; RRM: 76.2 vs. 76.9, p=0.45; WW: 83.7 vs. 82.9, p=0.08; zf-C2H2: 83.4 vs. 81.2, p=0.23; PDZ: 77.6 vs. 77.4, p=0.55; Pkinase: 88.7 vs. 89.7, p=0.11). In Kunitz, SA generation achieved significantly *higher* pLDDT than stored sequences (90.8 vs. 89.3, p=0.001, d=0.67). Only SH3 showed lower pLDDT for SA generation (70.5 vs. 74.1, p=0.006, d=−0.56), though the mean remained above the pLDDT=70 confidence threshold. Across all families, $\geq$ 86% of SA-generated sequences exceeded pLDDT=70 (with 100% in WW, Kunitz, and Pkinase).

TM-scores to family reference structures provided an even stronger result. In five of seven families, SA generation achieved significantly *higher* TM-scores than stored sequences (Wilcoxon $p < 0.001$; RRM: 0.409 vs. 0.380, d=1.62; WW: 0.181 vs. 0.174, d=1.14; Kunitz: 0.847 vs. 0.828, d=0.99;

PDZ: 0.638 vs. 0.574, d=1.41; Pkinase: 0.707 vs. 0.679, d=1.56). The remaining two families showed no significant difference (SH3: p=0.56; zf-C2H2: p=0.11). The large effect sizes ($d > 1.0$) in several families indicate that this is not merely a failure to reject the null: SA-generated sequences are structurally *more similar* to the canonical family fold than a random sample of natural family members. This likely reflects the Boltzmann ensemble's tendency to weight the central tendency of the family distribution, producing sequences closer to the "consensus fold" than individual natural sequences, which may contain lineage-specific structural deviations. The three families with uniformly low TM-scores (RRM, WW, zf-C2H2) showed the same pattern for stored sequences, indicating a length mismatch between the generated domain and the multi-domain reference structure rather than a failure of the generative method. To complement the aggregate statistics, we used AlphaFold2 (via ColabFold) to predict structures for the highest-confidence SA-generated sequence from each family and superimposed it on the experimentally determined reference structure (Fig. 6). The structural overlays confirm, at the level of individual folds, what the pLDDT and TM-score distributions indicate in aggregate: SA-generated sequences adopt the canonical architecture of their target family with sub-angstrom backbone RMSD, despite having no sequence identity to the reference and 42–65% novelty relative to all stored family members.

5 Discussion and Conclusion

We set out to ask what can be generated from a small protein family alignment with no training, and the answer is surprisingly rich: sequences that are novel, compositionally faithful, and structurally correct, produced in seconds on a laptop from nothing but the alignment itself. Across seven Pfam families spanning a ten-fold range of family sizes (K=37–420) and a seven-fold range of sequence lengths (L=23–262), stochastic attention matched or exceeded the output quality of methods pretrained on millions of sequences, while requiring zero external data, zero training, and zero GPU resources. The generated sequences preserve family-level amino acid composition (KL < 0.06), achieve substantial novelty (0.42–0.65), and fold into the correct three-dimensional architecture as confirmed by both ESMFold and AlphaFold2 structure prediction.

The structure validation results were particularly striking. Rather than merely matching the structural quality of natural sequences, SA-generated sequences achieved significantly higher TM-scores to the canonical family fold in five of seven families (Wilcoxon $p < 0.001$, Cohen's $d > 1.0$). This was not an artifact of sequence similarity: the same generated sequences had 42–65% novelty relative to stored patterns. The most likely explanation is that the Boltzmann ensemble defined by the modern Hopfield energy weights the central tendency of the family distribution, producing sequences that are closer to a "consensus fold" than individual natural sequences, which carry lineage-specific structural deviations accumulated over evolutionary time. The pLDDT confidence scores told a complementary story: SA-generated sequences were statistically indistinguishable from stored sequences in five of seven families ($p > 0.05$), confirming that the sampler does not sacrifice folding confidence for diversity. These results address the most immediate reviewer concern for any sequence-level generative method (do the sequences actually fold?) with a clear affirmative.

Comparison against three established baselines (profile HMM emit, EvoDiff, and the MSA Transformer) revealed a trade-off that stochastic attention uniquely resolves. All three baselines produced sequences with significantly lower identity to stored family members than SA generation ($p < 0.001$ in every family, with effect sizes $d = 1.6$–26.8). Profile HMM emit and the MSA Transformer generated sequences at 9–48% identity, well below the typical within-family range, suggesting sequences that have drifted outside the family's functional envelope. EvoDiff balanced the metrics more effectively but required pretraining on millions of UniRef50 MSAs and scaled prohibitively with sequence length ($\sim$18 hours for 150 Pkinase sequences on CPU, compared to seconds for SA). Stochastic attention occupied the middle ground (competitive KL divergence, high novelty, moderate sequence identity) using only the seed alignment. This makes the method particularly attractive for orphan families, newly discovered protein clades, or engineered libraries where external training corpora are unavailable or inappropriate.

The scaling study on the WW domain showed that this balance is robust: stochastic attention maintained stable generation quality down to K=20 sequences, with KL divergence below 0.02 and novelty above 0.47 even at this extreme. This robustness arises because the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set, and because the entropy inflection criterion automatically adapts the operating temperature to the geometry of the memory matrix at each scale. The critical temperature β^* itself can be predicted from PCA dimensionality alone (R^2=0.97), eliminating the need for a temperature sweep and enabling fully automatic operation from a seed alignment. In contrast, learned generative models face a fundamental parameter-to-data ratio problem when $K < 100$, a regime that encompasses the majority of Pfam families.

Several limitations should be acknowledged. Stochastic attention operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L, and cannot generate insertions, deletions, or variable-length sequences. The PCA projection discards information that may be relevant for capturing higher-order correlations, and the choice of retained variance (95%) involves a trade-off between compactness and expressiveness that we have not optimized. The multi-chain initialization strategy, while effective for the families studied here, does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families. Finally, our structure validation used ESMFold, which is itself a neural model with biases; wet-lab characterization of selected generated sequences would provide stronger evidence of functional viability and is a natural next step.

Looking beyond proteins, the connection between attention and Boltzmann sampling suggests a broader principle: any set of examples defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a principled generative model with no training. This perspective may extend to other domains where data is scarce but examples are structured, such as antibody repertoires, RNA families, or small-molecule libraries. The analytic score function also provides a natural interface with the convergence theory of Langevin dynamics, enabling formal guarantees on sample quality that are absent from most neural generative models.

In summary, stochastic attention is a competitive, training-free alternative to learned generative models for protein sequence generation, particularly in the small-family regime that characterizes most of protein space. The method requires only a seed alignment, has $\mathcal{O}(dK)$ per-step cost, needs no GPU, and produces structurally plausible sequences whose diversity and composition match, or exceed, those of natural family members. Code and data are available at `https://github.com/varnerlab/SA-Protein-Modeling-Study`.

Acknowledgments

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Author contributions. J.D.V. designed research, performed research, analyzed data, and wrote the paper.

Competing interests. The author declares no competing interest.

Data availability. All code, data, and analysis scripts are available at `https://github.com/varnerlab/SA-Protein-Modeling-Study`. Pfam seed alignments were obtained from InterPro [11]. Reference structures were obtained from the RCSB Protein Data Bank.

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Table 1: Properties of the seven Pfam families used in this study. K: number of seed sequences after cleaning. L: alignment length (columns retained). d: PCA dimension (95% variance). $\bar{H}_{\mathrm{col}}$: mean per-column Shannon entropy (nats). K_{eff}: effective number of sequences at 80% identity threshold. β^*: empirical entropy inflection point. $\beta^*/\sqrt{d}$: ratio to the random-pattern prediction.

Family	Pfam ID	K	L	d	$\bar{H}_{\mathrm{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55

Table 2: Generation metrics across seven Pfam families (mean $\pm$ SE). SA gen: stochastic attention generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination. KL: KL divergence of amino acid composition ($\downarrow$ lower is better; SE via 1,000-fold bootstrap). Novelty: $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space ($\uparrow$ higher is better). SeqID: nearest sequence identity to a stored pattern after decoding ($\downarrow$ lower is better, indicating greater novelty). Standard errors for novelty and SeqID are computed over 30 independent chains of 5 samples each.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$ ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	$\mathbf{0.060 \pm 0.005}$	$\mathbf{0.513 \pm 0.011}$	$\mathbf{0.538 \pm 0.006}$
	SA ret	0.107 ± 0.010	0.334 ± 0.020	0.616 ± 0.012
	BS	0.133 ± 0.006	0.000 ± 0.000	0.645 ± 0.004
	GP	0.132 ± 0.005	0.008 ± 0.000	0.633 ± 0.006
	CC	2.091 ± 0.000	0.510 ± 0.009	0.562 ± 0.000
SH3	SA gen	0.036 ± 0.004	$\mathbf{0.471 \pm 0.012}$	$\mathbf{0.593 \pm 0.006}$
	SA ret	0.035 ± 0.004	0.213 ± 0.012	0.722 ± 0.011
	BS	$\mathbf{0.030 \pm 0.003}$	0.000 ± 0.000	0.772 ± 0.006
	GP	$\mathbf{0.030 \pm 0.003}$	0.006 ± 0.000	0.752 ± 0.006
	CC	0.798 ± 0.012	0.474 ± 0.008	0.601 ± 0.002
WW	SA gen	$\mathbf{0.008 \pm 0.002}$	$\mathbf{0.653 \pm 0.006}$	$\mathbf{0.562 \pm 0.005}$
	SA ret	0.046 ± 0.013	0.354 ± 0.016	0.763 ± 0.011
	BS	0.024 ± 0.003	0.000 ± 0.000	0.823 ± 0.005
	GP	0.029 ± 0.004	0.017 ± 0.000	0.809 ± 0.006
	CC	1.375 ± 0.092	0.637 ± 0.004	0.766 ± 0.001
Kunitz	SA gen	$\mathbf{0.013 \pm 0.002}$	$\mathbf{0.557 \pm 0.009}$	$\mathbf{0.607 \pm 0.004}$
	SA ret	0.037 ± 0.003	0.257 ± 0.016	0.760 ± 0.012
	BS	0.023 ± 0.002	0.000 ± 0.000	0.834 ± 0.004
	GP	0.024 ± 0.002	0.010 ± 0.000	0.844 ± 0.004
	CC	0.721 ± 0.013	0.546 ± 0.007	0.590 ± 0.001
zf-C2H2	SA gen	0.013 ± 0.028	$\mathbf{0.596 \pm 0.010}$	$\mathbf{0.557 \pm 0.007}$
	SA ret	0.021 ± 0.010	0.167 ± 0.014	0.848 ± 0.013
	BS	$\mathbf{0.007 \pm 0.002}$	0.000 ± 0.000	0.938 ± 0.004
	GP	$\mathbf{0.007 \pm 0.002}$	0.011 ± 0.000	0.940 ± 0.004
	CC	2.346 ± 0.060	0.580 ± 0.006	0.600 ± 0.001
PDZ	SA gen	$\mathbf{0.038 \pm 0.010}$	$\mathbf{0.418 \pm 0.009}$	$\mathbf{0.543 \pm 0.005}$
	SA ret	0.059 ± 0.016	0.236 ± 0.017	0.626 ± 0.014
	BS	0.041 ± 0.002	0.000 ± 0.000	0.642 ± 0.009
	GP	0.045 ± 0.003	0.006 ± 0.000	0.648 ± 0.007
	CC	0.339 ± 0.001	0.449 ± 0.009	0.549 ± 0.001
Pkinase	SA gen	$\mathbf{0.035 \pm 0.001}$	$\mathbf{0.478 \pm 0.011}$	0.515 ± 0.004
	SA ret	0.050 ± 0.001	0.316 ± 0.013	0.531 ± 0.006
	BS	0.054 ± 0.001	0.000 ± 0.000	0.522 ± 0.003
	GP	0.051 ± 0.001	0.005 ± 0.000	0.533 ± 0.003
	CC	0.062 ± 0.001	0.446 ± 0.009	$\mathbf{0.470 \pm 0.000}$

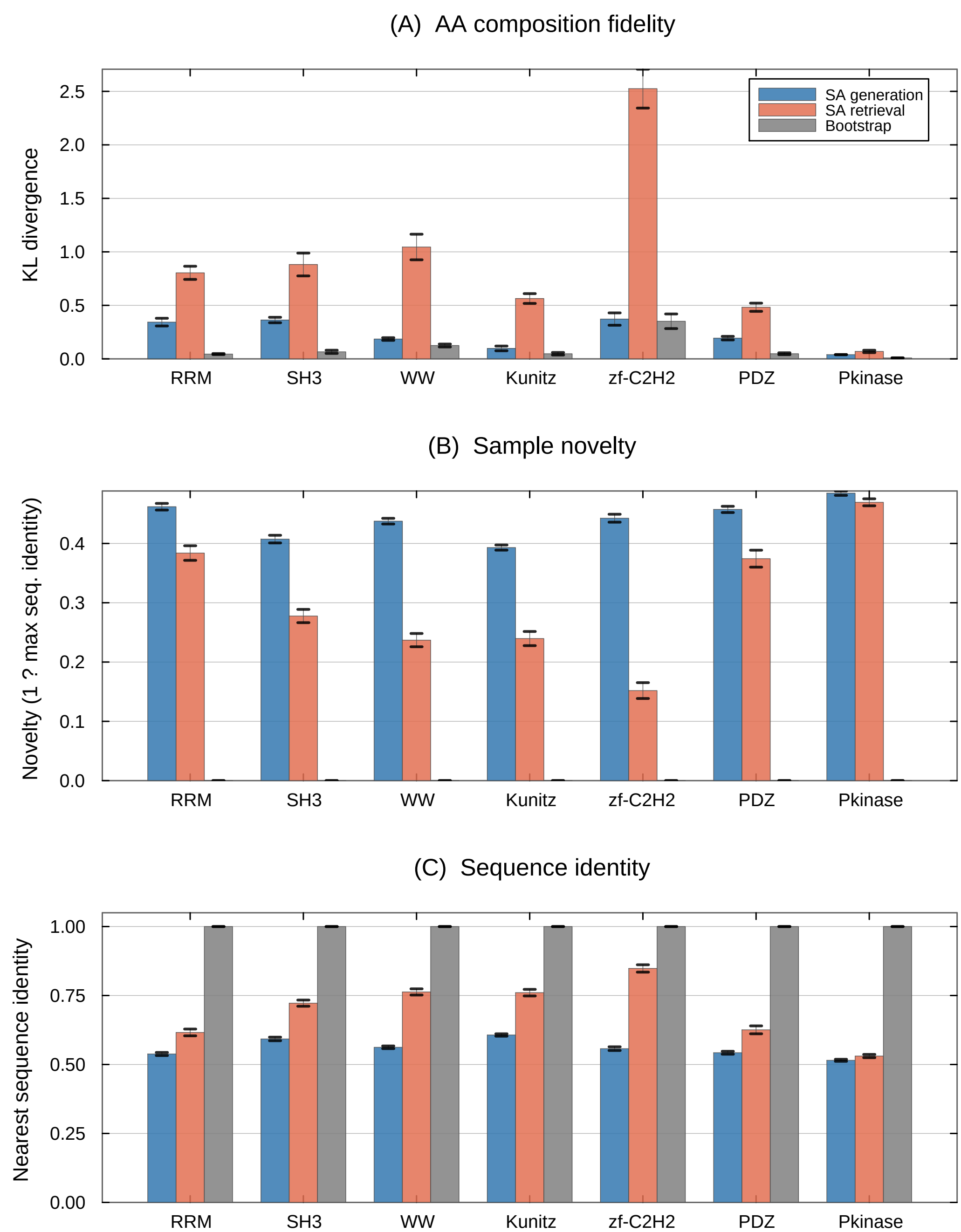

Figure 1: Cross-family comparison of SA generation, SA retrieval, and bootstrap across seven Pfam families. **(A)** KL divergence of amino acid composition (lower is better). SA generation achieves the lowest or near-lowest KL in every family. **(B)** Novelty in sequence space (1 $-$ max seq. identity; higher is better). SA generation produces substantially novel sequences, while bootstrap and Gaussian perturbation remain near zero. **(C)** Nearest sequence identity to a stored pattern (lower indicates greater departure). Error bars show ± 1 SE (30 chains $\times$ 5 samples).

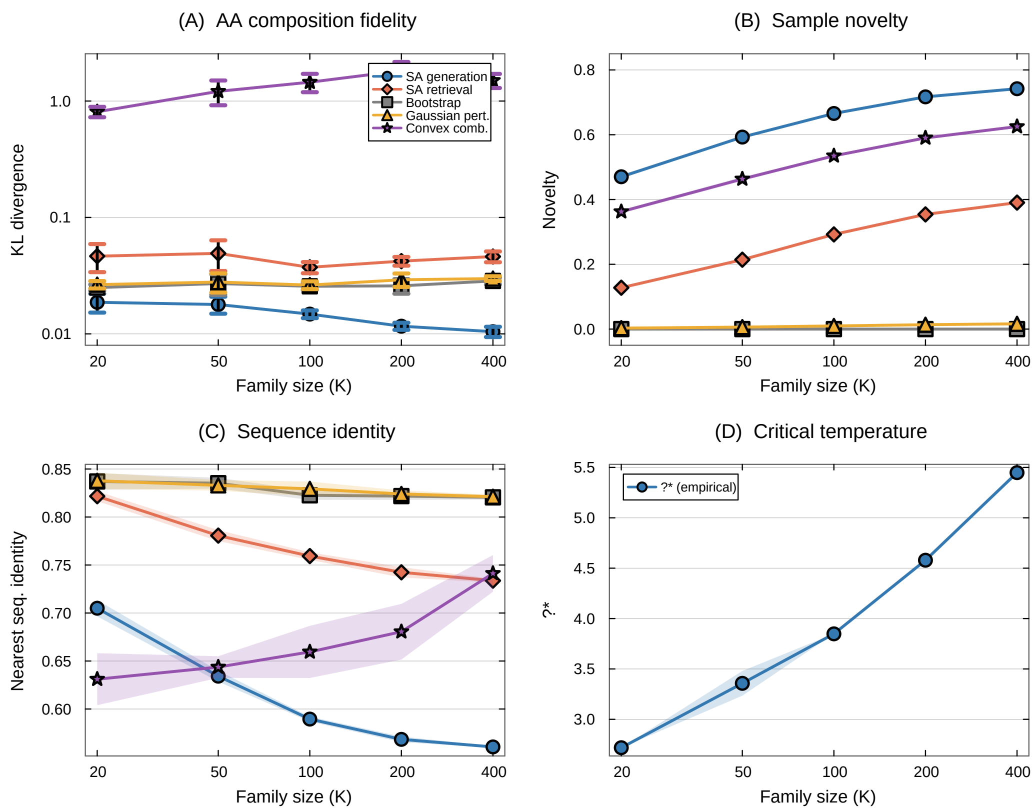

Figure 2: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log–log scale). SA generation maintains low KL across the entire range; convex combination degrades catastrophically. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K, indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

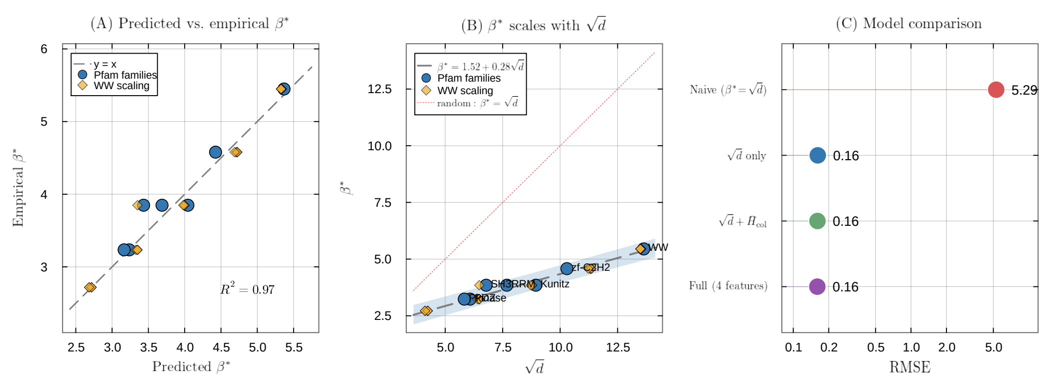

Figure 3: Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* for the simple linear model $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (R^2=0.97, n=32). Blue circles: seven Pfam families; gold diamonds: WW scaling replicates ($K \in \{20, \ldots, 400\}$, five repeats each). **(B)** β^* as a function of $\sqrt{d}$, with the fitted regression (dashed) and the random-pattern prediction $\beta^* = \sqrt{d}$ (dotted red). The intercept offset and reduced slope reflect structured correlations in real protein families. **(C)** Model comparison (log-scale RMSE). The simple $\sqrt{d}$ model (RMSE=0.16) is 33× more accurate than the naive random-pattern prediction $\beta^* = \sqrt{d}$ (RMSE=5.29); adding MSA features provides negligible improvement.

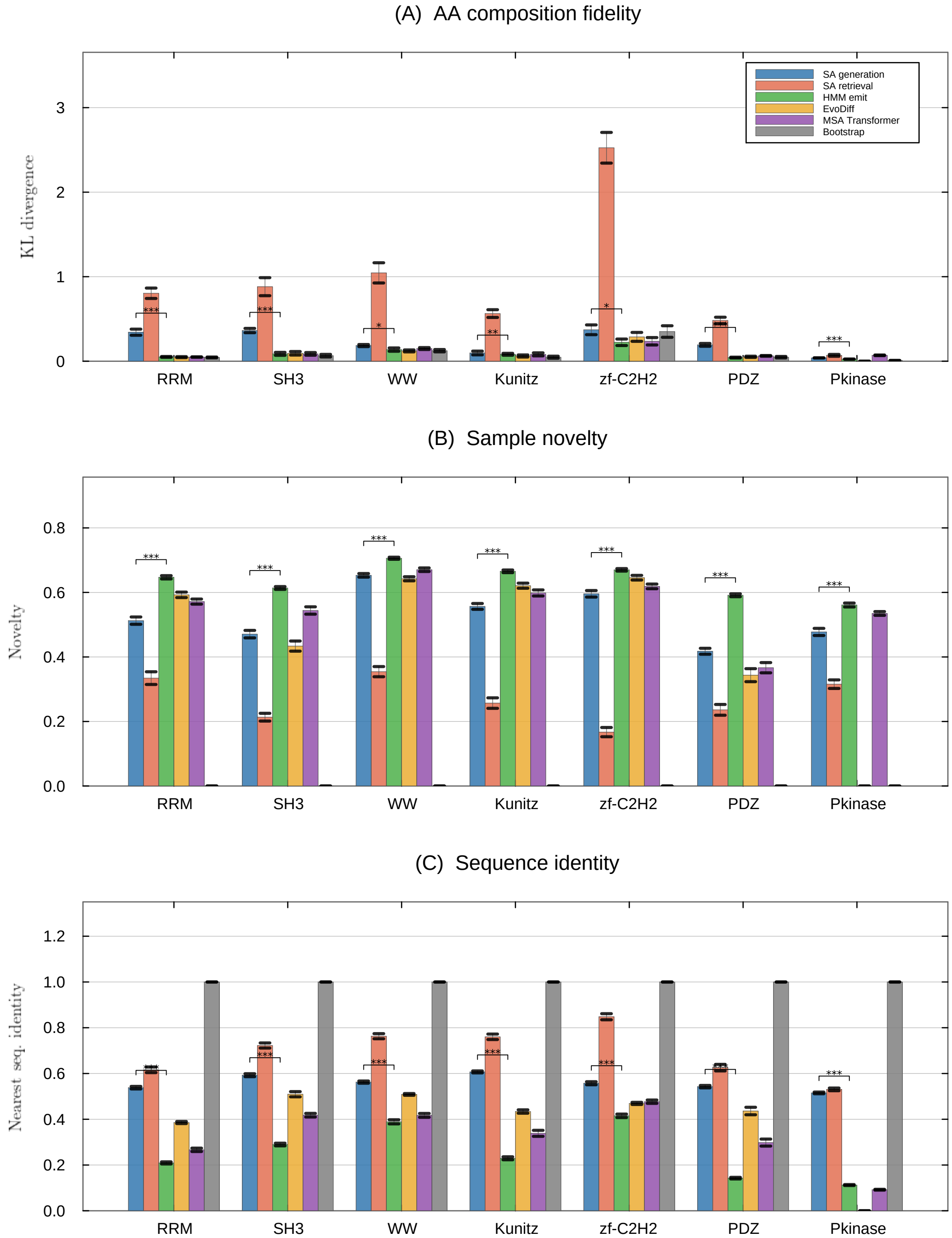

Figure 4: Head-to-head comparison of SA generation against learned baselines across seven Pfam families (six for EvoDiff, which could not complete Pkinase in feasible time). **(A)** KL divergence of amino acid composition. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity. Significance brackets show Wilcoxon rank-sum tests comparing SA generation to HMM emit (the most direct comparison, as both use only the seed alignment): $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, n.s. = not significant. Error bars show ± 1 SE (30 chains $\times$ 5 samples).

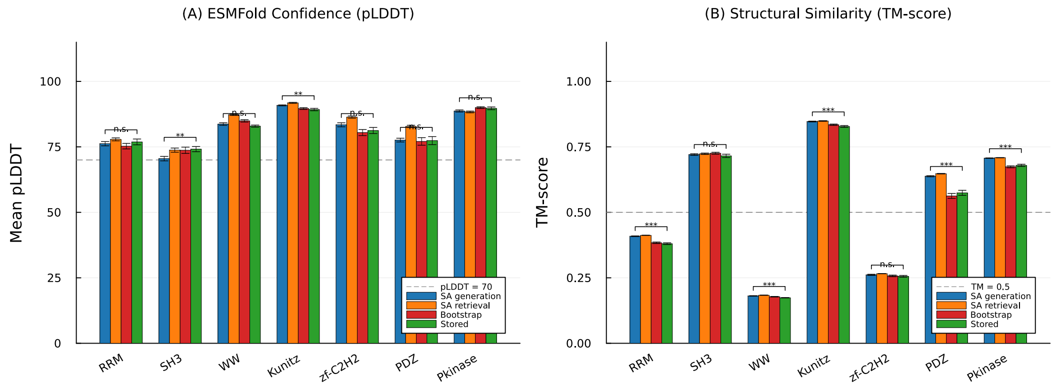

Figure 5: Structure validation of SA-generated sequences using ESMFold. **(A)** Mean pLDDT (folding confidence) across seven Pfam families for SA generation, SA retrieval, bootstrap, and stored (natural) sequences. The dashed line indicates the pLDDT=70 confidence threshold. **(B)** TM-score to a representative experimentally determined structure for each family. The dashed line indicates TM=0.5 (same-fold threshold). Significance brackets show Wilcoxon rank-sum tests comparing SA generation to stored sequences: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, n.s. = not significant. Error bars show ± 1 SE (n=50 sequences per condition).

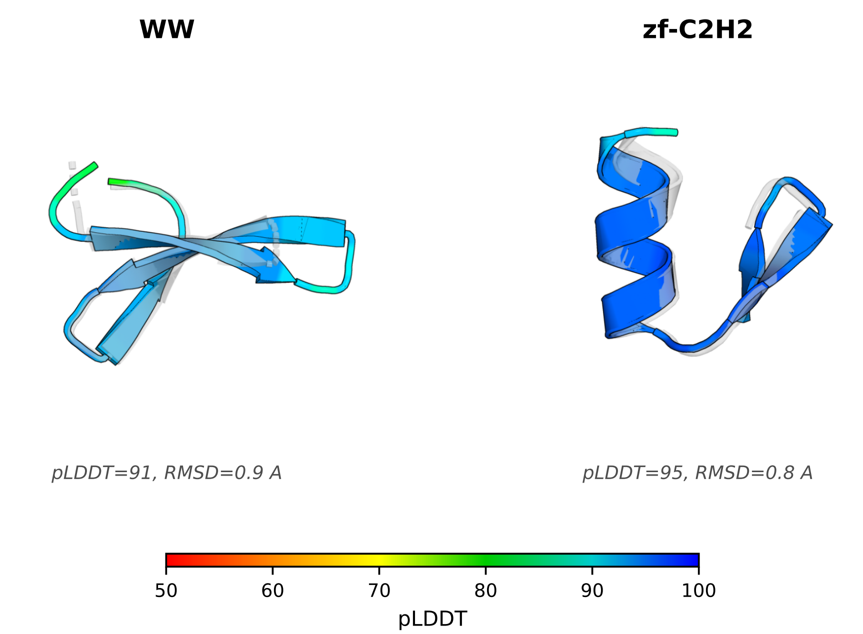

Figure 6: AlphaFold2 predicted structures for representative SA-generated sequences (colored by per-residue pLDDT confidence) superimposed on the experimentally determined reference structure for each family (gray, semi-transparent). For each family, the SA-generated sequence with the highest mean pLDDT was selected and aligned to the reference using PyMOL. The close structural agreement (sub-angstrom RMSD) and uniformly high pLDDT scores confirm that SA-generated sequences, despite substantial sequence-level novelty, encode three-dimensional folds that faithfully recapitulate the canonical architecture of their target family.

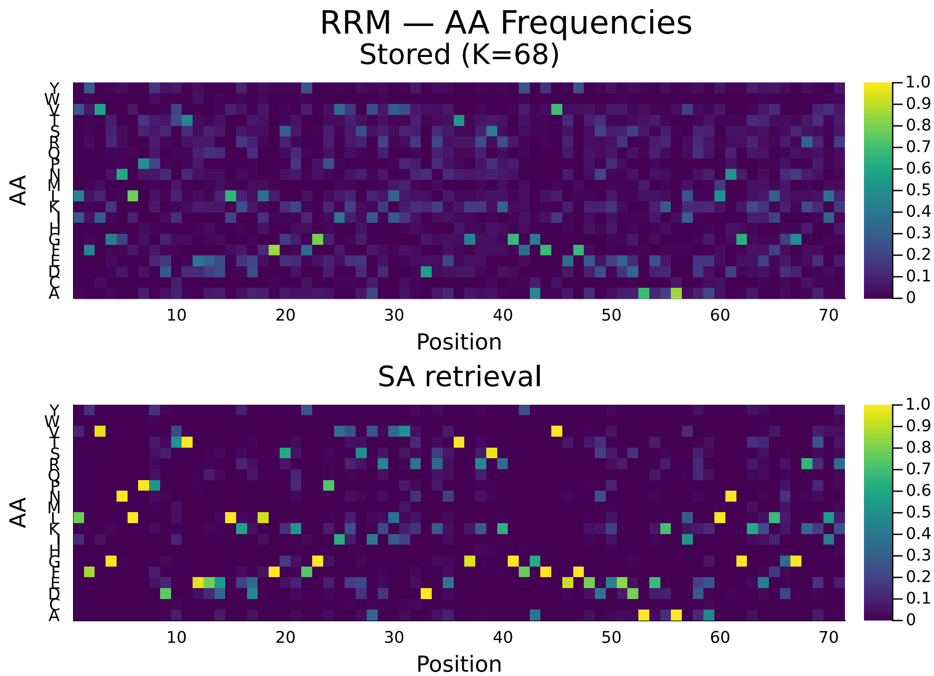

Figure S1: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences (K=68). **Bottom:** SA-generated sequences (S=150, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

SI Appendix

S1.1 Amino Acid Composition Profiles

To illustrate how stochastic attention preserves position-specific residue preferences, Fig. S1 compares the per-position amino acid frequency profiles of the stored seed sequences and SA-generated sequences for the RRM family (PF00076). The generated ensemble faithfully reproduces the conservation pattern of the family: strongly conserved positions (e.g., the aromatic residues characteristic of the RNP1 and RNP2 motifs) retain high frequency in the generated output, while variable positions display a similar breadth of amino acid usage. This agreement is consistent with the low KL divergences reported in Table 2 and confirms that the Boltzmann sampling procedure captures not only global amino acid composition but also the position-specific constraints encoded in the seed alignment.

S1.2 Analysis of the Critical Temperature

This section develops the analytical framework behind the empirical relationship $\beta^* \approx 1.57+0.28\sqrt{d}$ reported in Section 3, proceeding from exact results on the similarity variance through a Gaussian mean-field prediction to a diagnosis of why the mean-field coefficient overshoots and how stored-pattern self-similarity resolves the discrepancy.

S1.2.1 Similarity Variance and the Gaussian Inflection Baseline

After one-hot encoding and PCA projection to d dimensions, the stored patterns $\mathbf{m}_k$ are normalized to lie on $\mathbb{S}^{d-1}$. For a random query $\boldsymbol{\xi}$ drawn uniformly from $\mathbb{S}^{d-1}$, each similarity $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ has mean zero and variance $1/d$, regardless of the structure of $\mathbf{m}_k$. This follows from the rotational invariance of the uniform distribution on $\mathbb{S}^{d-1}$: for any fixed unit vector $\mathbf{m}$,

$$\mathbb{E}[\mathbf{m}^\top \boldsymbol{\xi}] = 0, \qquad \mathrm{Var}(\mathbf{m}^\top \boldsymbol{\xi}) = \frac{1}{d}. \tag{7}$$

We verified this numerically for all seven protein families using 1,000 random probes per family. The ratio $\sigma^2 \cdot d$ is 1.000 ± 0.003 across all families and WW subsamples (Fig. S2B,F), confirming that PCA-projected protein patterns behave identically to random unit vectors with respect to similarity variance. The excess kurtosis is mildly negative (-0.06 to -0.33), consistent with the sub-Gaussian behavior expected on $\mathbb{S}^{d-1}$. These observations motivate a Gaussian mean-field model: if the K similarity scores were i.i.d. draws from $\mathcal{N}(0, \sigma^2)$ with $\sigma = 1/\sqrt{d}$, one could work entirely in dimensionless units by writing each score as $e_k = \sigma z_k$ with $z_k \sim \mathcal{N}(0, 1)$.

In these units the softmax entropy $H(\tau) = -\sum_k p_k \log p_k$ with $p_k \propto \exp(\tau z_k)$ undergoes a transition from $H = \log K$ (uniform, $\tau = 0$) to $H \to 0$ ($\tau \to \infty$). We define the *dimensionless inflection* τ^* as the τ that maximizes $|d^2 H/d(\log \tau)^2|$ and compute it numerically by averaging over 500 Gaussian realizations for each $K \in \{10, 20, \ldots, 1000\}$. The result (Fig. S2A) is that τ^* is nearly constant at ≈ 1.6 for $K \geq 30$, with only weak logarithmic growth. Translating back to physical units, the Gaussian prediction for the entropy inflection is $\beta^* = \tau^*/\sigma = \tau^*\sqrt{d} \approx 1.6\sqrt{d}$, which has the correct $\sqrt{d}$ scaling and achieves a Pearson correlation of $r = 0.98$ with the empirical values across all families and subsamples.

S1.2.2 Discrepancy and the Role of Stored-Pattern Probes

Despite the strong correlation, the Gaussian prediction systematically overpredicts β^* by a factor of ≈ 4 ($R^2 = -180$), making it useless as an absolute predictor. The fitted relationship

$$\beta^* = 1.565 + 0.281/\sigma = 1.565 + 0.281\sqrt{d} \tag{8}$$

achieves $R^2 = 0.962$, with an intercept that matches τ^* to within numerical precision but a slope of 0.28, roughly $6\times$ smaller than $\tau^* \approx 1.6$. Understanding this gap requires examining how β^* is actually computed. The empirical inflection is evaluated by probing at *stored patterns* $\mathbf{m}_k$, not at random queries on $\mathbb{S}^{d-1}$. When $\boldsymbol{\xi} = \mathbf{m}_k$, the similarity vector acquires a distinctive structure:

$$e_j = \mathbf{m}_j^\top \mathbf{m}_k = \begin{cases} 1 & j = k \\ \text{cross-similarity} & j \neq k, \end{cases} \tag{9}$$

since $\|\mathbf{m}_k\| = 1$. The self-similarity score of 1 is an outlier relative to the cross-similarities $\mathbf{m}_j^\top \mathbf{m}_k$ for $j \neq k$, which have magnitudes $\mathcal{O}(1/\sqrt{d})$. The resulting gap $\Delta = 1 - \max_{j \neq k} \mathbf{m}_j^\top \mathbf{m}_k$, which ranges from 0.55 to 0.82 across families (Fig. S3E), means that the softmax concentrates on the self-pattern at much lower β than would be needed for a random query where no score is pinned at the maximum of $\mathbb{S}^{d-1}$.

We investigated whether the Gaussian τ^*/σ framework could be salvaged by substituting the cross-similarity standard deviation σ_{cross} for $\sigma = 1/\sqrt{d}$, but this fails decisively (Fig. S3D; $R^2 = -9.0$). The cross-similarity variance does not follow the $1/d$ law (it ranges from $0.19\times$ to $2.4\times$ the random-probe variance depending on the family's correlation structure; Fig. S3B,C), and the excess kurtosis is large (6–25), reflecting a heavy-tailed distribution driven by a few highly similar pattern pairs amid a bulk of near-orthogonal ones. The Gaussian mean-field decomposition $\beta^* = \tau^*/\sigma$ thus breaks down completely for stored-pattern probes: the self-similarity outlier and the non-Gaussian cross-similarity tails violate both assumptions simultaneously. Nonetheless, the fitted model $\beta^* = 1.57 + 0.28\sqrt{d}$ remains robust. The $\sqrt{d}$ scaling is set by concentration of measure, which holds for any probe, and the self-similarity effect enters primarily through a reduced effective coefficient rather than a change in functional form. The intercept $\approx \tau^*$ recovers the universal softmax baseline, while the coefficient 0.28 encapsulates the combined effect of self-similarity anchoring and the protein family correlation structure, a quantity that is empirically stable across the seven families and five scaling conditions studied here, but for which we do not currently have a closed-form expression.

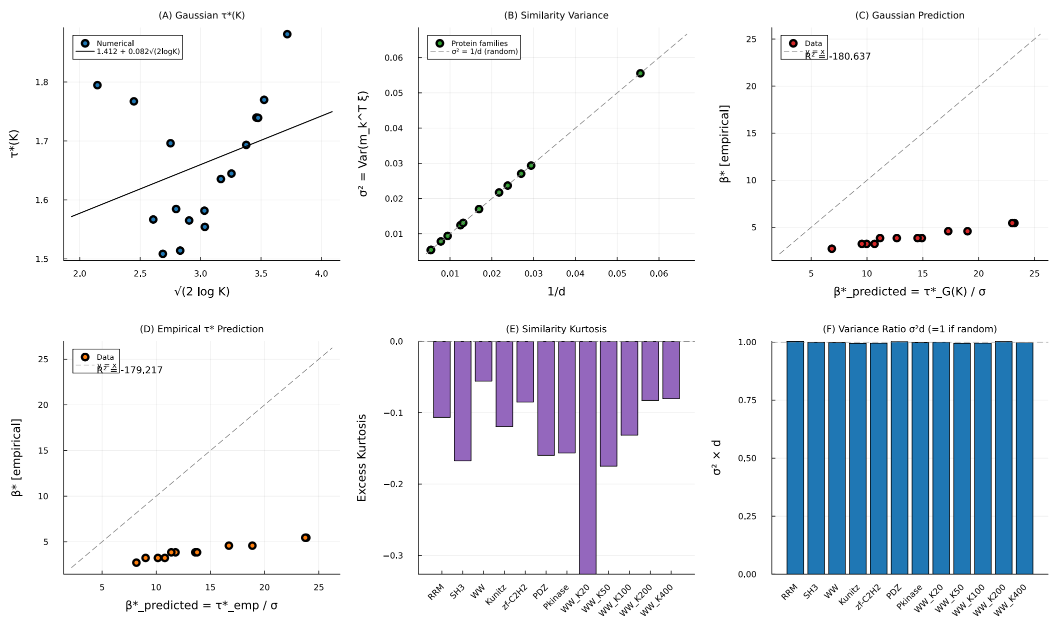

Figure S2: **Analytical investigation of β^*.** (A) Dimensionless softmax inflection $\tau^*(K)$ for Gaussian scores; the dashed line is a linear fit in $\sqrt{2\log K}$. (B) Similarity variance $\sigma^2 = \mathrm{Var}(\mathbf{m}_k^\top \boldsymbol{\xi})$ vs. $1/d$ for random probes; all points lie on the identity line, confirming $\sigma^2 = 1/d$. (C) Gaussian prediction $\beta^*_{\text{pred}} = \tau^*_G(K)/\sigma$ vs. empirical β^*; the prediction has the correct trend ($r = 0.98$) but overshoots by $\sim 4\times$. (D) Same as (C) using the empirical (non-Gaussian) τ^*; the overshoot persists. (E) Excess kurtosis of the similarity distribution across families. (F) The ratio $\sigma^2 d$ across all datasets, confirming the concentration-of-measure prediction $\sigma^2 d = 1$.

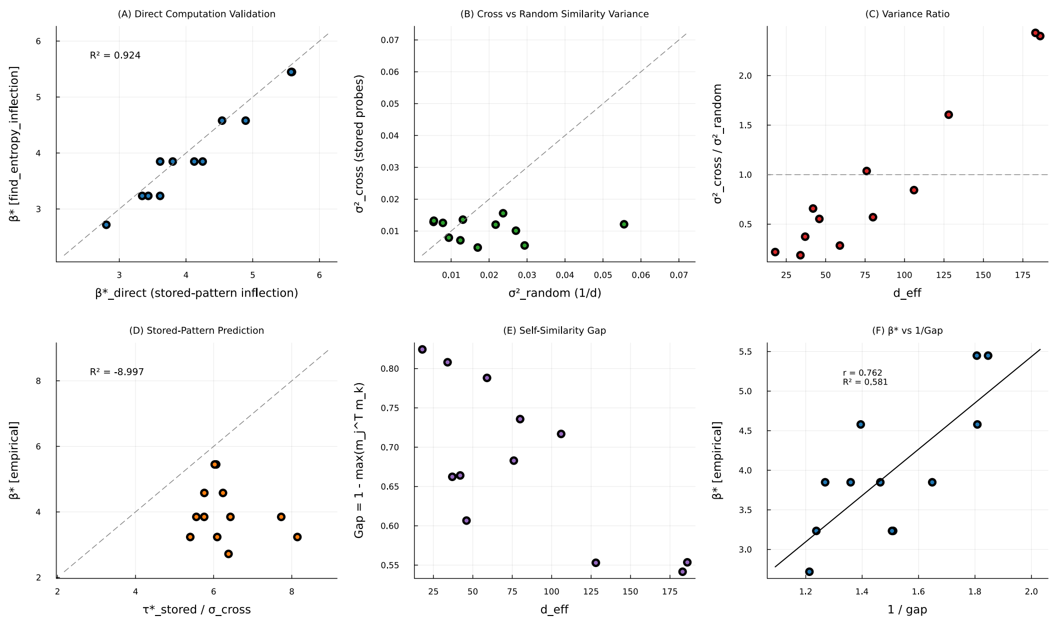

Figure S3: **Stored-pattern probe analysis.** (A) Validation: β^* recomputed from stored-pattern inflection vs. the value from `find_entropy_inflection` ($R^2 = 0.924$). (B) Cross-similarity variance (stored probes) vs. random-probe variance ($1/d$); stored probes systematically deviate from the identity line. (C) Variance ratio $\sigma^2_{\text{cross}}/\sigma^2_{\text{random}}$ vs. d; no consistent relationship. (D) The decomposition $\tau^*_{\text{stored}}/\sigma_{\text{cross}}$ fails as a predictor of β^* ($R^2 = -9.0$). (E) Self-similarity gap $\Delta = 1 - \max_{j \neq k} \mathbf{m}_j^\top \mathbf{m}_k$ vs. d. (F) β^* vs. $1/\Delta$ ($r = 0.76$); the gap has moderate predictive power but is noisier than $\sqrt{d}$.